

```
const int trigPin = 7;  
const int echoPin = 6;  
const int greenLED = 8;  
const int yellowLED = 9;  
const int redLED = 10;
```

```
const int potPin = A0;
```

```
void setup()  
{  
  pinMode(trigPin, OUTPUT);  
  pinMode(echoPin, INPUT);  
  pinMode(greenLED, OUTPUT);  
  pinMode(yellowLED, OUTPUT);  
  pinMode(redLED, OUTPUT);
```

```
  Serial.begin(9600);  
}
```

```
void loop()  
{  
  int potValue = analogRead(potPin);  
  int warningDistance = map(potValue, 0, 1023, 10, 50);
```

```
  digitalWrite(trigPin, LOW);  
  delay(100);
```

```
  digitalWrite(trigPin, HIGH);
```

```
delayMicroseconds(10);  
digitalWrite(trigPin, LOW);  
  
long duration = pulseIn(echoPin, HIGH);  
  
float distance = duration * 0.034 / 2;  
  
if (distance > warningDistance)  
{  
    digitalWrite(greenLED, HIGH);  
    digitalWrite(yellowLED, LOW);  
    digitalWrite(redLED, LOW);  
  
    Serial.println("Status: SAFE");  
}  
  
else if (distance > warningDistance / 2)  
{  
    digitalWrite(greenLED, LOW);  
    digitalWrite(yellowLED, HIGH);  
    digitalWrite(redLED, LOW);  
  
    Serial.println("Status: GETTING CLOSE");  
}  
  
else  
{  
    digitalWrite(greenLED, LOW);
```

```
digitalWrite(yellowLED, LOW);
```

```
digitalWrite(redLED, HIGH);
```

```
Serial.println("Status: DANGER");
```

```
}
```

```
Serial.print("Distance: ");
```

```
Serial.print(distance);
```

```
Serial.println(" cm");
```

```
Serial.print("Warning Distance: ");
```

```
Serial.print(warningDistance);
```

```
Serial.println(" cm");
```

```
delay(500);
```

```
}
```

